

Mark schemes

Q1.

Marking Instructions	AO	Marks	Typical Solution
States integral(s) of the cubic with limits that include 1 and 4	AO1.1a	M1	$\text{mean} = k = \frac{1}{4-1} \int_1^4 (x^3 - 7x^2 + 11x + 6) dx$
Integrates the function and substitutes correct limits. Condone one incorrect term. $\int_1^4 (x^3 - 7x^2 + 11x + 6) dx = \frac{69}{4}$ Note: $\Rightarrow$ M1M1	AO1.1a	M1	$\therefore k = \frac{1}{3} \left[\frac{x^4}{4} - \frac{7x^3}{3} + \frac{11x^2}{2} + 6x \right]_1^4$ $= \frac{1}{3} \left(\frac{4^4}{4} - \frac{7 \times 4^3}{3} + \frac{11 \times 4^2}{2} + 6 \times 4 \right)$ $- \frac{1}{3} \left(\frac{1^4}{4} - \frac{7 \times 1^3}{3} + \frac{11 \times 1^2}{2} + 6 \times 1 \right)$
Obtains the correct value of k Do not apply ISW. NMS can score 3/3.	AO1.1b	A1	$= \frac{1}{3} \times \frac{80}{3} - \frac{1}{3} \times \frac{113}{12}$ $= 5.75$

$\int_1^4 (x^3 - 7x^2 + 11x + 6 - k) dx = 0$ $\left[\frac{x^4}{4} - \frac{7x^3}{3} + \frac{11x^2}{2} + 6x - kx \right]_1^4 = 0$ $\left(\frac{4^4}{4} - \frac{7 \times 4^3}{3} + \frac{11 \times 4^2}{2} + 6 \times 4 - k \times 4 \right)$ $- \left(\frac{1^4}{4} - \frac{7}{3} + \frac{11}{2} + 6 - k \right) = 0$ $64 - \frac{448}{3} + 88 + 24 - 4k - \frac{113}{12} + k = 0$ $\frac{69}{4} = 3k$ $k = 5.75$
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Area B = Area C



$$\begin{aligned}
 &\therefore \int_1^p (x^3 - 7x^2 + 11x + 6 - k) dx \\
 &= \int_p^4 (k - x^3 + 7x^2 - 11x - 6) dx \\
 &= \left[\frac{x^4}{4} - \frac{7x^3}{3} + \frac{11x^2}{2} + 6x - kx \right]_1^p \\
 &= \left[kx - \frac{x^4}{4} + \frac{7x^3}{3} - \frac{11x^2}{2} - 6x \right]_p^4 \\
 &= \left(\frac{p^4}{4} - \frac{7p^3}{3} + \frac{11p^2}{2} + 6p - kp \right) \\
 &\quad - \left(\frac{1}{4} - \frac{7}{3} + \frac{11}{2} + 6 - k \right) \\
 &= \left(4k - 64 + \frac{7 \times 64}{3} - \frac{11 \times 16}{2} - 24 \right) \\
 &\quad - \left(kp - \frac{p^4}{4} + \frac{7p^3}{3} - \frac{11p^2}{2} - 6p \right) \\
 &= -\frac{1}{4} + \frac{7}{3} - \frac{11}{2} - 6 + k \\
 &= 4k - 64 + \frac{448}{3} - 88 - 24 \\
 &\frac{69}{4} = 3k \\
 &k = 5.75
 \end{aligned}$$

Total 3 marks

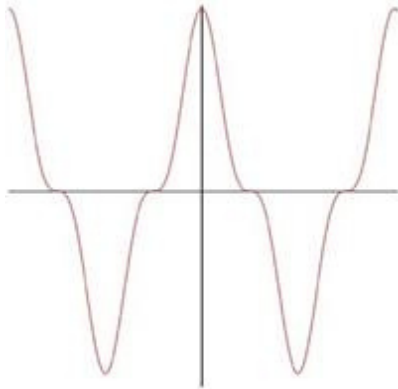
Q2.

Marking Instructions	AO	Marks	Typical Solution
Circles correct answer	AO1.1b	B1	13
Total 1 mark			

Q3.

Marking Instructions	AO	Marks	Typical Solution
Correctly manipulates $y = [f(x)]^n$ into a form that can be sketched.	AO3.1a	M1	$\cosh(ix) \frac{e^{ix} + e^{-ix}}{2}$
Sketches an appropriate section of the graph.	AO1.1b	M1	$= \frac{\cos(x) + i \sin(x) + \cos(-x) + i \sin(-x)}{2}$



Sketch does not need to be accurate but should look symmetrical about y-axis and the area above and below the x-axis should look similar			$= \frac{\cos(x) + t \sin(x) + \cos(x) - t \sin(x)}{2}$ $= \cos(x)$ $\therefore [f(x)]_n = \cos n(x)$ 
Deduces, with fully correct reasoning, the correct mean value	AO2.2a	R1	Since the graph has the same shape/area above and below the x-axis over the given domain, the mean value of $y = [f(x)]_n$ must be 0.
Total 3 marks			

Examiner reports

Q1.

Only a minority of students realised that this question was asking for the mean value. A common approach was to translate the graph k units downwards and then solve

$\int_1^4 f(x) = 0$. It was not uncommon for students to assume that the intersection with $y = k$ occurred at $x = 2.5$. However, it was possible to use the unknown intersection point and solve $\int_p^4 y \, dx = -\int_1^p y \, dx$. Unfortunately, many students who attempted this method struggled with the subsequent algebra.